

Result 1.

Hydrodynamic entropy of GraphCast: weight energy on the mesh lengths in the checkpoint

1 What is being measured

If E_i is the energy at spatial scale i ,

$$p_i = \frac{E_i}{\sum_j E_j}, \quad H = - \sum_i p_i \ln p_i, \quad H_n = \frac{H}{\ln N}.$$

H is in nats. H_n divides that entropy by its maximum, $\ln N$, the value H takes when all N scales hold equal energy. The wavenumber of a level is $k = 1/L$, with L the physical length of that level, and the reference slope on the log-log plot is Kolmogorov's $-5/3$.

The lengths are the icosahedral levels stored in each checkpoint. Level m has characteristic length

$$L(m) = \frac{\pi R}{2 \cdot 2^m}, \quad R = 6371 \text{ km}.$$

GraphCast Small has `mesh_size` = 5, so the levels are M_0 – M_5 , from 10008 km to 313 km. The operational and full checkpoints have `mesh_size` = 6, so they add M_6 at 156 km. The shorter rungs in a ten-level illustration, near 50 km, 25 km, and 12 km, are absent from these checkpoints and are left out of the sum.

2 Where the energy comes from

GraphCast keeps one mesh-edge encoder and applies it to every edge of the multimesh. The checkpoint therefore has no separate weight matrix for M_0 , M_1 , and so on. The sum of squared entries of each matrix is a single number, the same at every length. For the three published checkpoints that shared sum is about 10 for the first matrix. The second matrix is larger and differs by checkpoint: 152 for Small, 35 for operational, and 137 for the full model.

A spectrum against $1/L$ can still be formed, because those shared weights act on the edges of each real length. An edge carries the four geometric numbers the encoder was trained to read,

$$\mathbf{x}_e = (\Delta x, \Delta y, \Delta z, \|\Delta\|),$$

the displacement between its endpoints on the unit sphere and the length of that displacement. The encoder then does three things, and each one is reported:

1. **linear_0**. The first matrix only, $W_0 \in \mathbb{R}^{4 \times 512}$,

$$\mathbf{y}_e = \mathbf{x}_e W_0 + \mathbf{b}_0.$$

This is the quantity computed in `Main codes`.

2. **mlp.** Both linear maps. A swish sits between them, and the second matrix is $W_1 \in \mathbb{R}^{512 \times 512}$.
3. **encoder.** The same vector after the trained layer norm, which is the edge embedding this encoder writes. The norm removes the mean and the variance of the 512 components of each edge and then applies a learned scale and offset.

The energy of a level is the mean, over the edges of that level, of the squared Euclidean norm of the chosen output. Dividing by the edge count keeps $E(m)$ from simply counting edges: each refinement multiplies the number of edges by four. The probabilities, H , and the fit of $E(k)$ use only the levels the checkpoint contains.

Nothing is retrained. No ERA5 field is loaded. The processor edge maps take a 1536-dimensional input, the sender node, the edge, and the receiver node together, so they are not evaluated on edge geometry alone.

3 How this differs from the notebooks in Main codes

	Notebooks in Main codes	This calculation
Weights	<code>linear_0</code> of the mesh-edge encoder	Both linear maps, then the trained layer norm
Mesh lengths	M_0 – M_5 or M_0 – M_6 , from the checkpoint	The same lengths. Shorter rungs are left out
$E(m)$	Mean $\ W_0 \mathbf{x} + \mathbf{b}_0\ ^2$	Mean squared output of the stage above
Entropy	Reported in bits	Reported in nats, with $H_n = H / \ln N$
Wavenumber	$k = 1/L(m)$	$k = 1/L(m)$

The first row of each block below reproduces those notebooks in nats. For Small, 0.835 nats is their 1.205 bits, and $H_n = 0.466$ is unchanged by the change of base.

4 Results

Model	Stage	H (nats)	H_n	slope	share on M_0
Small	<code>linear_0</code>	0.835	0.466	−1.866	70.5%
	both linear maps	0.847	0.473	−1.777	70.2%
	after layer norm	1.789	0.998	−0.061	17.8%
Operational	<code>linear_0</code>	0.836	0.429	−1.846	70.5%
	both linear maps	0.842	0.433	−1.773	70.3%
	after layer norm	1.912	0.983	−0.187	17.3%
Full	<code>linear_0</code>	0.836	0.430	−1.843	70.5%
	both linear maps	0.833	0.428	−1.889	70.6%
	after layer norm	1.922	0.988	−0.154	16.7%

$\ln 6 = 1.792$ and $\ln 7 = 1.946$. The fit is $\ln E$ against $\ln k$ with $k = 1/L$ in km^{-1} . For the first matrix the coefficient of determination is 0.999 on Small and 0.999 on the two seven-level models. After the layer norm it falls to 0.83, 0.79, and 0.76.

On GraphCast Small the three energies, in the same order of stages, are:

Level	L (km)	edges	E , linear_0	E , both maps	E , layer norm
M_0	10008	60	4.579	26.77	4156
M_1	5004	240	1.411	8.279	4133
M_2	2502	960	0.375	2.238	4098
M_3	1251	3840	0.0965	0.602	3977
M_4	625	15360	0.0256	0.176	3666
M_5	313	61440	0.0078	0.0631	3344

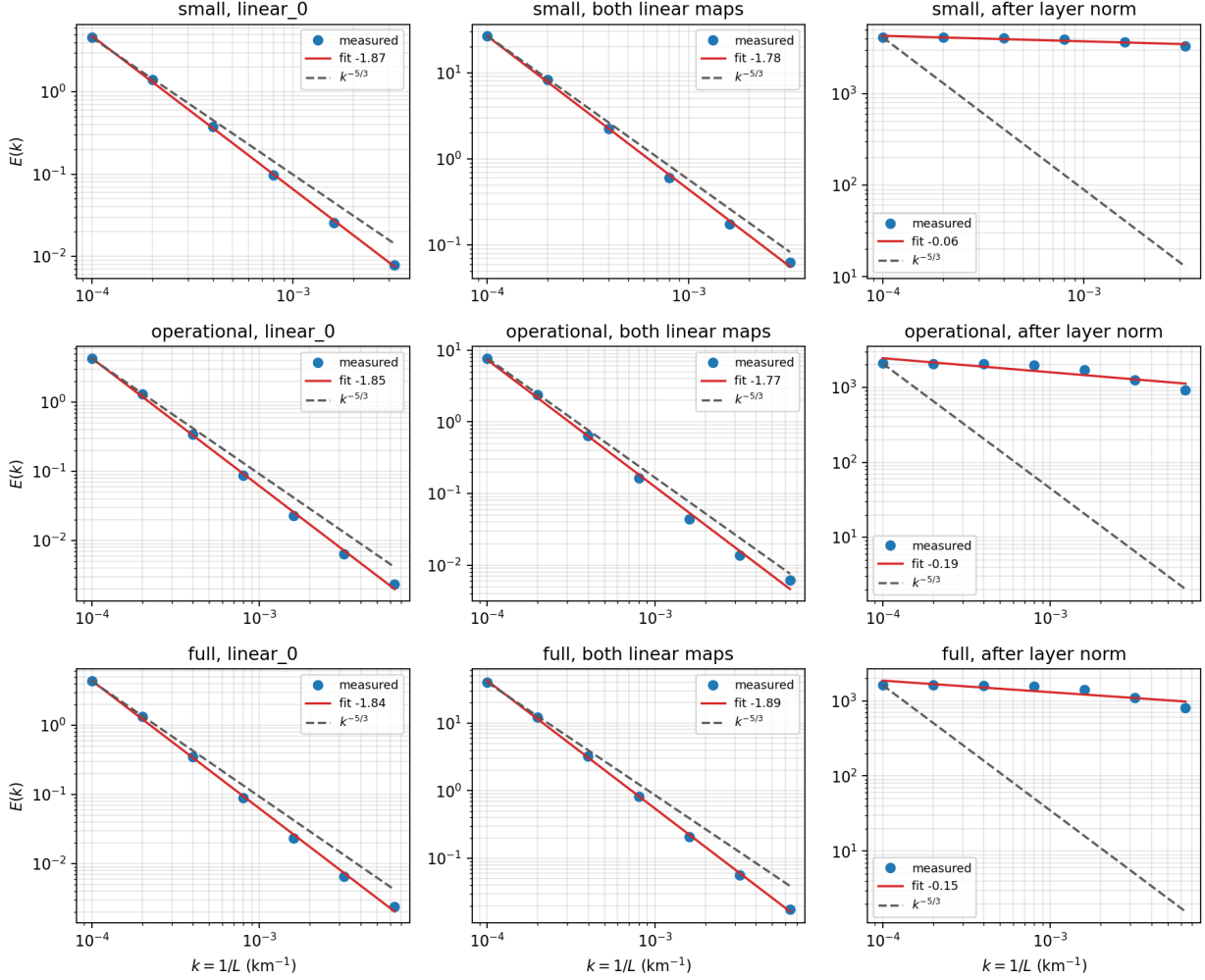


Figure 1: Energy against $k = 1/L$. Each panel is one checkpoint and one stage. Circles are the measured energy. The solid line is the power-law fit of that panel, with its slope in the legend. The dashed line is $k^{-5/3}$, drawn through that panel's own M_0 energy. Absolute height is not comparable across columns.

5 Why the number changes

The 0.835 nats of the first matrix is fixed by the mesh. A long edge has a larger input than a short one, and $\|\mathbf{x}\|^2$ scales as length squared. Length halves at each refinement, so $E(m)$ falls by

about a factor of four and the slope sits near -2 . The fitted -1.87 , with almost all of the variance explained, is that geometric scaling after one linear map. About 70% of the energy lies on M_0 . The small, operational, and full checkpoints agree on this slope and on this share, which is what a shared geometric input should do.

Passing that vector through the second weight matrix does not remove the scaling. Swish is applied to a vector whose size still tracks the edge length, and the second matrix is shared across levels, so the slope only moves from -1.87 to about -1.78 on Small and stays near -1.8 on the other two checkpoints. H_n stays near 0.47 for six levels and near 0.43 for seven.

The layer norm is the step that changes the entropy. It divides out the magnitude of each edge, and the magnitude is where the length entered. What remains is the direction of the 512-vector, rescaled by the learned scale and offset. On Small that leaves $H = 1.789$ nats and $H_n = 0.998$, with 17.8% of the energy on M_0 against an equal share of 16.7%. The operational and full checkpoints, which have a seventh level, land at $H_n = 0.983$ and 0.987. The residual slope is shallow, -0.06 on Small and about -0.19 to -0.15 when M_6 is included. That is a property of these weights on the lengths they actually have. It is a different object from an illustration that places energy on ten lengths down to about 12 km and quotes an exponent near $-1/2$.